GL BAJAJ
Institute of Technology & Management
FIND YOUR SPARK

Department of Computer Science & Engineering

TITLE OF MINI PROJECT

MINI PROJECT | SESSION 2026-2027

PRESENTED BY :
TEAM NO. :
STUDENT NAME(S) : S1,S2, S3, S4
ROLL NO(S) : R1,R2, R3, R4
BRANCH & SECTION :
YEAR : 2nd Year (B.Tech)

UNDER SUPERVISION OF :
PROJECT GUIDE :
DESIGNATION :

GL BAJAJ
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OUTLINE

01 Introduction

02 Objectives & Scope

03 Existing vs Proposed System

04 Technology Stack

05 Resource Requirements

06 System Design & Architecture

07 Methodology

08 Modules & Features

09 Implementation & Results

10 Testing & Validation

11 Conclusion & Future Scope

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FORMAT & GUIDELINES

Content Format

- Heading: 32 pt | Subheading: 18 pt
- Paragraph / bullet text: 16 pt
- Use one bullet style across all slides
- Running text must be justified
- Short points, not long paragraphs

Presentation Rules

- Maximum 15 slides, in the given order
- Every team member presents one part
- Keep a working demo ready, with screenshots as backup
- Add your GitHub repository link
- Cite every source used (APA style)

INTRODUCTION

1 **Domain & Background**

Name the domain (e.g., Web Development, Machine Learning, Cyber Security) and explain the basic idea behind it in 2-3 lines.

2 **Problem Statement**

State the real-world problem your project solves, who faces it, and why the current approach is not enough.

3 **Motivation**

Explain why you chose this project, who the target users are, and what benefit it brings.

OBJECTIVES & SCOPE

Objectives

Write 3-4 clear goals, each starting with an action verb.

- To design and develop a working application for the stated problem
- To implement secure login and role-based access
- To store and retrieve data using a database
- To test the system and present a working demo

Scope

Be honest about what this version does and does not cover.

Included

- Core features delivered in this version
- Target users and platforms

Not included (limitations)

- Features planned for future versions
- Known constraints (data, time, hosting)

Tip: link every objective to a result shown in Implementation & Results.

EXISTING vs PROPOSED SYSTEM

Existing System

- Manual, paper-based or scattered tools
- Time-consuming and error-prone
- No central data or user access control

Proposed System

- Automated, web-based solution
- Central database with fast search and reports
- Secure login and a responsive interface

Literature / similar applications reviewed: name 2-3 existing apps, papers or tutorials and note one gap each leaves open.

TECHNOLOGY STACK

1 Frontend

- HTML5, CSS3, JavaScript
- React.js / Bootstrap / Tailwind

2 Backend

- Node.js + Express, Django or Flask
- REST APIs, JWT authentication

3 Database & Tools

- MongoDB / MySQL / PostgreSQL
- Git & GitHub, VS Code, Postman

ML / Data Science projects

Python, NumPy, Pandas, scikit-learn, Matplotlib, Jupyter Notebook

Other basic projects

Python, Java or C++ with a GUI, or Android basics

RESOURCE REQUIREMENTS

Hardware (minimum)

Item	Specification
Processor	Intel Core i3 / Ryzen 3 or above
RAM	8 GB (4 GB minimum)
Storage	256 GB SSD / HDD or more
Network	Stable internet connection

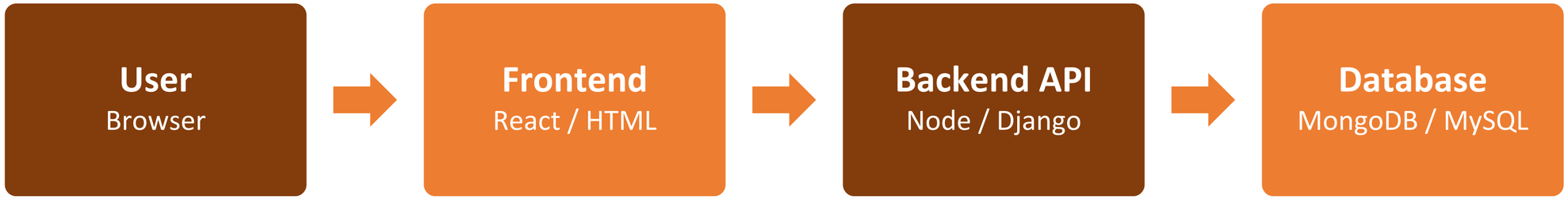
Write the exact versions you actually used for your project.

Software & Tools

Item	Specification
OS	Windows 10/11 or Ubuntu
IDE / Editor	VS Code, PyCharm
Runtime	Node.js / Python 3.x
Database	MongoDB / MySQL
Version control	Git & GitHub
Browser	Chrome / Edge

SYSTEM DESIGN & ARCHITECTURE

Typical 3-tier architecture (replace with your own)



Insert your diagram here
Architecture, ER or DFD

Include

- Use case diagram
- DFD (Level 0 / 1)
- ER diagram
- Flowchart or ML pipeline

METHODOLOGY

1

Requirement Analysis
Collect needs and fix the feature list

2

Design
Plan screens, database and flow

3

Development
Build frontend, backend and database

4

Testing
Check features and fix bugs

5

Deployment
Host the app and prepare the demo

Project timeline: mention the duration of each phase and the model followed (Agile / Waterfall).

MODULES & FEATURES

01

User Authentication
Register, log in and keep sessions secure

02

Dashboard
Overview of user activity and data

03

Core Feature (CRUD)
Create, view, update and delete records

04

Search & Filter
Find the right data quickly

05

Admin Panel
Manage users and content

06

Reports & Alerts
Export data and send notifications

IMPLEMENTATION & RESULTS

Paste screenshot

Login / Home screen

Paste screenshot

Main feature screen

Paste screenshot

Output / Result screen

GitHub link: add your repository URL. **ML projects:** show accuracy, confusion matrix and key graphs instead of UI screens.

TESTING & VALIDATION

Test case	Input	Expected result	Status
Valid login	Correct email, password	Dashboard opens	Pass
Invalid login	Wrong password	Error message shown	Pass
Add new record	Valid form data	Record saved in database	Pass
Model accuracy (ML)	Test dataset	Accuracy above target	Pass

Testing done: unit, functional and integration testing, plus browser and device checks. For ML projects, use a train-test split with accuracy, precision and recall.

CONCLUSION & FUTURE SCOPE

1 Conclusion

- Summarise what was built and achieved
- State how each objective was met
- Mention key learning: teamwork, tools and debugging

2 Future Scope

- Add new features such as payments and notifications
- Build a mobile app or PWA version
- Deploy on cloud and add analytics or AI

Key takeaway: close with one sentence on what the project achieved and what you learned.

REFERENCES

[1] Author, A. A., & Author, B. B. (Year). Title of the article. Journal Name, Volume(Issue), pages.
[2] MDN Web Docs. (2026). HTTP overview. Mozilla. https://developer.mozilla.org
[3] scikit-learn developers. (2026). User guide. https://scikit-learn.org

Use APA style. List only the documentation, papers and tutorials you actually used.

Thank You

Any questions?